Linear Algebra*

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1 Euclidean Space

- $\mathbb{R}$: set of all real numbers.
- $\mathbb{N}$: set of all natural numbers, i.e., $\mathbb{N} = \{1, 2, 3, \cdots, \}$.
- $\mathbb{Z}$: set of all integers, i.e., $\mathbb{Z} = \{x | x = 0 \text{ or } x \in \mathbb{N} \text{ or } -x \in \mathbb{N}\}$.

For any set X, we can form **n-tuples**, or **lists**, of elements of X. We write an n-tuple as $(x_1, x_2, \cdots, x_n)$, where each **component** x_i is an element of X. Note that the order matters, i.e., $(2, 5, 2)$ and $(5, 2, 2)$ are distinct, different 3-tuples of numbers. Therefore, they are sometimes called as **ordered** lists or ordered n-tuples.

Note that n-tuple and a set are different. The order of a set's elements does not matter and elements cannot repeat. For example, a set $\{2, 5, 2\}$ does not make sense and the sets $\{2, 5\}$ and $\{5, 2\}$ are the same set. For any given set X, we use the notation X^n to denote the set of all n-tuples of elements of X. So, $\mathbb{R}^n$ is the set of all n-tuples $(x_1, \cdots, x_n)$, where each component x_i is a real number. Similarly, $\mathbb{Z}^n$ is the set of all n-tuples of integers and $\mathbb{N}^n$ is the set of all n-tuples of natural numbers. For example, 3-tuples $(2, 5, 2)$ and $(5, 2, 2)$ are elements of $\mathbb{N}^3$, $\mathbb{Z}^3$ and $\mathbb{R}^3$.

The geometry with the sets $\mathbb{R}^n$ is also important. Geometrically, the set $\mathbb{R}$ is identified "real line", a line with a designated point that represents the number 0 and where each number $x \in \mathbb{R}$ is represented by the point $|x|$ units of distance to the right of 0 (if $x > 0$) or to the left of 0 (if $x < 0$). The set $\mathbb{R}^2$ is represented geometrically as "the Euclidean plane"

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and the elements of $\mathbb{R}^2$ as points on the plane, where the two components x_1 and x_2 of an ordered pair (2-tuple) $(x_1, x_2) \in \mathbb{R}^2$ are the **Cartesian coordinates** of the point associated with the pair (x_1, x_2) in $\mathbb{R}^2$ as the point (x_1, x_2). Similarly, the set $\mathbb{R}^3$ is represented as a three-dimensional space and the elements (x_1, x_2, x_3) of $\mathbb{R}^3$ as points whose Cartesian coordinates are the components x_1, x_2 and x_3. As in $\mathbb{R}^2$, we often refer to the elements (x_1, x_2, x_3) of $\mathbb{R}^3$ as points in $\mathbb{R}^3$. Generalizing this to any natural number n, we refer to $\mathbb{R}^n$ as n-dimensional space and its elements $(x_1, x_2, \cdots, x_n)$ as points in $\mathbb{R}^n$, even though for $n > 3$ we cannot actually draw a diagram of $\mathbb{R}^n$.

For the notation, n-tuples will be denoted in boldface as $\mathbf{x} = (x_1, \cdots, x_n) \in \mathbb{R}^n$. Note that the elements of $\mathbb{R}^n$ are called points, but they are also called **vectors**. Focusing on $\mathbb{R}^2$, $\mathbf{x} = (x_1, x_2) \in \mathbb{R}^2$ can be interpreted as a *position* or a *location* in $\mathbb{R}^2$-plane. We can also interpret (x_1, x_2) as a *movement* or *displacement* from one location on the plane to another.

2 Vectors and Vector Spaces

2.1 Vector spaces

Definition 2.1. A triple $(V, +, \cdot)$ where

- set V, whose elements are called **vectors**.
- an operation $+$: $V^2 \to V$ called vector **addition** (if $\mathbf{u}$ and $\mathbf{v}$ are vectors, we note $\mathbf{u} + \mathbf{v}$).
- an operation $\cdot$: $\mathbb{R} \times V \to V$ called **scalar multiplication** (if $\mathbf{v}$ is a vector and λ a real, we note $\lambda \mathbf{v}$).

is said to be a (real) **vector space** (or a vector space over $\mathbb{R}$) iff it satisfies the following 7 axioms:

(1) Vector addition is commutative and associative: $\forall \mathbf{u}, \mathbf{v}, \mathbf{w} \in V$, $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ and $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$.

(2) Existence of an identity element of addition: there exists an **zero vector** noted $\mathbf{0}$ st. $\mathbf{0} + \mathbf{u} \; = \mathbf{u} + \mathbf{0} = u, \forall \mathbf{u} \in V$.

(3) Existence of an inverse element of addition: for any $\mathbf{u} \in V$, there exists an **additive inverse** of $\mathbf{u}$, noted $-\mathbf{u}$, st. $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$.

(4) Existence of an identity element of scalar multiplication: 1, a multiplicative identity in $\mathbb{R}$, st. $1\mathbf{u} = \mathbf{u}$.

(5) Mixed associativity: for any $\lambda_1, \lambda_2 \in \mathbb{R}$ and $\mathbf{v} \in V$, we have $(\lambda_1\lambda_2)\mathbf{v} = \lambda_1(\lambda_2\mathbf{v})$.

(6) Scalar multiplication is distributive w.r.t. vector addition: $\lambda(\mathbf{v}_1 + \mathbf{v}_2) = \lambda\mathbf{v}_1 + \lambda\mathbf{v}_2$.

(7) Scalar multiplication is distributive w.r.t. addition in $\mathbb{R}$: $(\lambda_1 + \lambda_2)\mathbf{v} = \lambda_1\mathbf{v} + \lambda_2\mathbf{v}$.

Exercise 1. *Prove the following results using the 7 axioms of a vector space:*

1. *The zero vector $\boldsymbol{0}$ is unique in a vector space.*
2. *The additive inverse of vector $\boldsymbol{v} \in V$ is unique.*
3. *$-\boldsymbol{v} = (-1)\boldsymbol{v}$. And therefore we can define vector subtraction by $\boldsymbol{v}_1 - \boldsymbol{v}_2 := \boldsymbol{v}_1 + (-\boldsymbol{v}_2)$.*
4. *$0\boldsymbol{v} = \boldsymbol{0}, \forall \boldsymbol{v}$; $\lambda\boldsymbol{0} = \boldsymbol{0}, \forall\lambda \in \mathbb{R}$, and that $\lambda\boldsymbol{v} = 0$ implies that either $\lambda = 0$ or $\boldsymbol{v} = \boldsymbol{0}$.*

A major example of a n-dimensional real vector space is $< \mathbb{R}^n, +, \cdot >$, where the vector addition and scalar multiplication are defined in a component-by-component fashion. An element $\mathbf{v}$ in $\mathbb{R}^n$ is $\mathbf{v} = (v_1, v_2, \cdots, v_n)$, where $v_i \in \mathbb{R}$ are called **components** or **coordinates** of $\mathbf{v}$. The zero vector is the vector whose components are all zero.

For a vector space $(V, +\cdot)$, a subset W is called a **vector subspace** of $(V, +\cdot)$ iff $(W, +|_W, \cdot|_W)$ is a vector space, where $+|_W$ and $\cdot|_W$ are $+$ and $\cdot$ defined for V restricted in W.

Definition 2.2. $(W, +, \cdot)$ is a vector subspace of $(V, +, \cdot)$ iff:

(1) W contains the zero vector $\mathbf{0}$.

(2) W is **closed** under vector addition: $\forall \mathbf{u}, \mathbf{v} \in V, \mathbf{u} + \mathbf{v} \in W$.

(3) W is **closed** under scalar multiplication: $\forall \mathbf{u} \in W, \forall\lambda \in \mathbb{R}, \lambda\mathbf{u} \in W$.

the conditions (1) and (2) can be replaced by the following one: $\lambda\mathbf{u} + \mu\mathbf{v} \in W, \forall\lambda, \mu \in \mathbb{R}, \forall \mathbf{u}, \mathbf{v} \in V$.

In order to determine whether a nonempty subset of a vector space V is actually a subspace of V, it is sufficient to merely check whether it is closed under vector addition and scalar multiplication. If it is, then its operations, under which V is a vector space, must necessarily satisfy 7 axioms.

Definition 2.3. Let $(V, +, \cdot)$ be a vector space and S a subset of V. The **linear span of** S, noted $\text{span}(S)$, is the smallest vector subspace that contain S, that is the intersection of all the subspace that contain S:

$$span(S) = \bigcap\{W, W \text{ a vector subspace of } V, \text{ and } S \subseteq W\}$$

Because intersection of vector subspaces is still a vector subspace, $span(S)$ is a vector subspace, which we call the **vector subspace spanned (or generated) by** S.

Definition 2.4. Let $\mathbf{v}_1, \cdots, \mathbf{v}_n$ be n vectors of V. A **linear combination** of $\mathbf{v}_1, \cdots, \mathbf{v}_n$ is a vector $\lambda_1\mathbf{v}_1 + \cdots + \lambda_n\mathbf{v}_n$ for n scalars $\lambda_1, \cdots, \lambda_n \in \mathbb{R}$.

It is straightforward that for a finite subset S of V, $span(S)$ is simply the set of all linear combinations of vectors in S. Note that while the set S can be infinite, every linear combination of members of S must be a linear combination of only a finite members of S, according to the definition of linear combination.

2.2 Linear Dependence

Definition 2.5. In vector space $(V, +, \cdot)$, a finite set of vectors $\mathbf{v}_1, \cdots, \mathbf{v}_n$ are said to be **linearly independent**, iff the linear combination $\sum_{i=1}^{n} \lambda_i\mathbf{v}_i = \mathbf{0}$ implies $\lambda_i = 0$ for any i. Otherwise, $\mathbf{v}_1, \cdots, \mathbf{v}_n$ are said to be linearly dependent.

Clearly, if a vector can be represented by a linear combination of a set of linearly independence vector, then the representation is unique.

Proposition 2.6. Let $\mathbf{v}_1, \cdots, \mathbf{v}_n$ be linearly independent elements of a vector space $(V, +, \cdot)$. Let $(\lambda_i)_i$ and $(\mu_i)_i$ be reals. If:

$$\lambda_1\mathbf{v}_1 + \cdots + \lambda_n\mathbf{v}_n = \mu_1\mathbf{v}_1 + \cdots + \mu_n\mathbf{v}_n,$$

then for all i, $\lambda_i = \mu_i$.

Proposition 2.7. In a vector space, $\mathbf{v}_1, \cdots, \mathbf{v}_n$ are linearly dependent iff $\exists\mathbf{v}_i (i \in \{1, 2, \cdots, n\})$ that can be represented by a linear combination of $(\mathbf{v}_j)_{j\neq i}$.

Theorem 2.8. *In vector space $(V, +, \cdot)$, if $\boldsymbol{u}_1, \cdots, \boldsymbol{u}_m$ can be represented by linear combinations of $\boldsymbol{v}_1, \cdots, \boldsymbol{v}_n$, and $m > n$, then $\boldsymbol{u}_1, \cdots, \boldsymbol{u}_m$ are linearly dependent.*

Shortly, if "more" can be represented by "less", then "more" must be linearly dependent.

2.3 Basis

Definition 2.9. For a set S in a vector space $(V, +, \cdot)$, the vectors $\mathbf{v}_1, \cdots, \mathbf{v}_n \in S$ are a **(finite) basis** of S, iff

(1) $\mathbf{v}_1, \cdots, \mathbf{v}_n$ are linearly independent, and

(2) All vectors in S can be represented by linear combinations of $\mathbf{v}_1, \cdots, \mathbf{v}_n$.

In other words, the basis of S is linearly independent subset of S that spans S. Note that because of condition (1), we know that the representation in condition (2) is unique. Note that the basis of S is not unique. However, two bases must have the same number of vectors. To see this, suppose $\mathbf{u}_1, \cdots, \mathbf{u}_m$ and $\mathbf{v}_1, \cdots, \mathbf{v}_n$ are both bases of S and $m < n$. Then, $\mathbf{u}_1, \cdots, \mathbf{u}_m$ can represent $\mathbf{v}_1, \cdots, \mathbf{v}_n$ and therefore $\mathbf{v}_1, \cdots, \mathbf{v}_n$ are linearly dependent by Theorem 2.8. This contradicts the assumption that $\mathbf{v}_1, \cdots, \mathbf{v}_n$ are a basis of S. Therefore, it is without ambiguity to define the **rank** of $S \subset V$ as the number of vectors in its basis, denoted as $rank(S)$. This also gives next theorem.

Theorem 2.10. *Let $\mathcal{B}$ be a finite basis of a vector space V. Any set $S \subseteq V$ containing more vectors than $\mathcal{B}$ is linearly dependent.*

Proposition 2.11. In vector space $(V, +, \cdot)$, $\mathbf{v}_1, \cdots, \mathbf{v}_n$ are a basis of $S \subset V$ iff they have the maximum number of linearly independent vectors in S.

Proof. ($\Leftarrow$): Take any $\mathbf{y} \in S$. Claim that $\mathbf{y}$ can be represented by a linear combinations of $\mathbf{v}_1, \cdots, \mathbf{v}_n$. Assume that $\mathbf{v}_1, \cdots, \mathbf{v}_n$ have the maximum number of linearly independent vectors in S. Then, the vectors $\mathbf{v}_1, \cdots, \mathbf{v}_n$ and $\mathbf{y}$ are linearly dependent. Then, we can write as:

$$\sum_{i=1}^{n} \lambda_i \mathbf{v}_i + \mu \mathbf{y} = \mathbf{0} \ (\mu \neq 0)$$

Thus, we can re-write as

$$\mathbf{y} = \sum_{i=1}^{n} \Big(-\frac{\lambda_i}{\mu} \Big) \mathbf{v}_i$$

and this tells us that $\mathbf{y}$ is a linear combinations of $\mathbf{v}_1, \cdots, \mathbf{v}_n$.

($\Rightarrow$): Suppose there exists a set of linearly independent vectors $\mathbf{u}_1, \cdots, \mathbf{u}_m$ in S with $m > n$. Since $\mathbf{v}_1, \cdots, \mathbf{v}_n$ are a basis of S, the vectors $\mathbf{v}_1, \cdots, \mathbf{v}_n$ can represent $\mathbf{u}_1, \cdots, \mathbf{u}_m$ and therefore, $\mathbf{u}_1, \cdots, \mathbf{u}_n$ are linearly dependent. (contradiction) $\square$

Proposition 2.12. In vector space $(V, +, \cdot)$, $\mathbf{v}_1, \cdots, \mathbf{v}_n$ are a basis of $S \subset V$ iff they have the minimum number of vectors in S that can represent all vectors in S as their linear combinations.

Proof. ($\Leftarrow$): Suppose that $\mathbf{v}_1, \cdots, \mathbf{v}_n$ are not linearly independent. Then we can represent one of them as a linear combination of the others. WLOG, suppose $\mathbf{v}_n$ can be represented by $\mathbf{v}_1, \cdots, \mathbf{v}_{n-1}$. Because all vectors in S can be represented by $\mathbf{v}_1, \cdots, \mathbf{v}_n$, they can also be represented by $\mathbf{v}_1, \cdots, \mathbf{v}_{n-1}$. This contradicts the assumption that $\mathbf{v}_1, \cdots, \mathbf{v}_n$ have the minimum number of vectors in S.

($\Rightarrow$): Suppose there exists a set of vectors, $\mathbf{u}_1, \cdots, \mathbf{u}_m$ in S that can represent all vectors in S and $m < n$. Then, $\mathbf{u}_1, \cdots, \mathbf{u}_m$ can represent $\mathbf{v}_1, \cdots, \mathbf{v}_n$ and therefore, $\mathbf{v}_1, \cdots, \mathbf{v}_n$ are linearly dependent. This contradicts that $\mathbf{v}_1, \cdots, \mathbf{v}_n$ are a basis of S. $\square$

Proposition 2.13. In vector subspace $(V, +, \cdot)$, suppose $\mathbf{v}_1, \cdots, \mathbf{v}_n$ are a basis. Then, $span(S) = span(\{\mathbf{v}_1, \cdots, \mathbf{v}_n\})$ of $S \subset V$.

If we let $S = V$, we can talk about the basis and the rank of the whole vector space V. Then, the rank of the whole space V is usually called the **dimension** of the vector space V and denoted as $\dim(V)$. If $\dim(V) = n$, then we say that the vector space is n-**dimensional**. Clearly, any set of more than n vectors in an n-dimensional vector space must be linearly dependent. For example, the space $\mathbb{P}_n$ of all polynomials of degree at most n, consisting of all polynomials of the form

$$p(t) = a_0 + a_1 t + \cdots + a_n t^n;$$

is an $n+1$-dimensional vector space, for which one basis is $\{1, t, \cdots, t^n\}$. It is also possible that a vector space V does not have a finite basis, in which case we say that V is **infinite-dimensional**. In an n-dimensional vector space, it can be shown that $\mathbf{v}_1, \cdots, \mathbf{v}_n$ are a basis of V if they are linearly independent, or they can represent all vectors in V. The **canonical basis** of the vector space $\mathbb{R}^n$ is $\mathbf{e}_1, \cdots, \mathbf{e}_n$, where $\mathbf{e} \coloneqq (1, 0, 0, \cdots, 0)$, $\mathbf{e}_2 \coloneqq (0, 1, 0, \cdots, 0)$, and so forth. Therefore, $\dim(\mathbb{R}^n) = n$.

2.4 Inner Products, Norms and Metrics

Definition 2.14. The **absolute value** of a number $x \in \mathbb{R}$ is $|x| \coloneqq \max\{x, -x\}$, which we also call it as size, magnitude, length, or norm of x.

Proposition 2.15. The absolute value function $|\cdot| : \mathbb{R} \to \mathbb{R}_+$ satisfies the following properties:

(1) $\forall x \in \mathbb{R} : |x| \geq 0$,

(2) $\forall x \in \mathbb{R} : |x| = 0 \Leftrightarrow x = 0$,

(3) $\forall x, y \in \mathbb{R} : |x + y| \leq |x| + |y|$,

(4) $\forall \alpha \in \mathbb{R}, x \in \mathbb{R} : |\alpha x| = |\alpha||x|$.

Definition 2.16. The **distance** between two numbers x and x' in $\mathbb{R}$ is the absolute value of their difference: $d(x, x') \coloneqq |x - x'|$.

Proposition 2.17. The distance function $d : \mathbb{R} \times \mathbb{R} \to \mathbb{R}_+$ satisfies the following properties:

(1) $d(x, x') \geq 0$,

(2) $d(x, x') = 0 \Leftrightarrow x = x'$,

(3) $d(x, x') = d(x', x)$,

(4) $d(x, x'') \leq d(x, x') + d(x', x'')$ (**triangle inequality**)

Definition 2.18. Let X be a set. A **metric** on X is a function $d : X \times X \to \mathbb{R}_+$ that satisfies Proposition 2.17 (1) - (4). The pair (X, d) is called a **metric space**.

Now, we move our focus to the length of a vector. As in $\mathbb{R}$, the distance between two points is the norm of their difference.

Definition 2.19. Let $(V, +, \cdot)$ be a vector space. The 4-tuple $(V, +, \cdot, < \cdot, \cdot >)$ is an **inner product space** iff the inner product operator $\langle \cdot, \cdot \rangle V \times V \to \mathbb{R}$ satisfies the following properties:

(1) Commutativity: $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$ for any $\mathbf{u}, \mathbf{v} \in V$,

(2) Linearity: $\langle \lambda_1 \mathbf{u}_1 + \lambda_2 \mathbf{u}_2, \mathbf{v} \rangle = \lambda_1 \langle \mathbf{u}_1, \mathbf{v} \rangle + \lambda_2 \langle \mathbf{u}_2, \mathbf{v} \rangle$ for any $\lambda_1, \lambda_2 \in \mathbb{R}$ and $\mathbf{u}_1, \mathbf{u}_2, \mathbf{v} \in V$, and

(3) Positive definiteness: $\langle \mathbf{u}, \mathbf{u} \rangle \geq 0$ for any $\mathbf{u} \in V$, and equality holds iff $\mathbf{u} = 0$.

Note that linearity also implies $\langle \mathbf{v}, \mathbf{0} \rangle = 0$ for any $\mathbf{v} \in V$, because $\langle \mathbf{v}, \mathbf{0} \rangle = \langle \mathbf{v}, 0\mathbf{v} \rangle = 0 \cdot \langle \mathbf{v}, \mathbf{u} \rangle = 0$ where $\mathbf{u}$ is an arbitrary vector in V. In an inner product space $(V, +, \cdot, \langle \cdot, \cdot \rangle)$, two vectors $\mathbf{v}$ and $\mathbf{u}$ are said to be **orthogonal** iff $\langle \mathbf{u}, \mathbf{v} \rangle = 0$. A leading example of inner products is the **dot product** defined on $\mathbb{R}^n$. The dot product of two vectors $\mathbf{x} = (x_1, \cdots, x_n)$ and $\mathbf{y} = (y_1, \cdots, y_n)$ in $\mathbb{R}^n$ is defined as

$$\mathbf{x} \cdot \mathbf{y} \coloneqq \sum_{i=1}^{n} x_i y_i.$$

Note that the dot product for two vectors is different from the scalar multiplication, which is defined for a scalar and a vector, although we usually use the same notation, $\cdot$. It is straightforward to verify that the dot product satisfies our requirements on inner products.

An inner product induces a **norm** $\|\cdot\| : V \to \mathbb{R}_+$ by

$$\|\mathbf{v}\| := \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}.$$

Using the properties of inner product, it is straightforward to show that (1) $\|\mathbf{v}\| = 0$ iff $\mathbf{v} = \mathbf{0}$, and (2) $\|\lambda \mathbf{v}\| = |\lambda| \|\mathbf{v}\|$. In $\mathbb{R}^n$, the norm induced by the dot product

$$\|\mathbf{x}\|_2 := \sqrt{\sum_{i=1}^{n} x_i^2}$$

is called the **Euclidean norm**, or L_2 **norm**.

Exercise 2. *Assume* $\|(x_1, 0, 0)\| = |x_1|$, $\|(0, x_2, 0)\| = |x_2|$ *and* $\|(0, 0, x_3)\| = |x_3|$. *Using Pythagorean Theorem, justify*

$$\|(x_1, x_2, x_3)\|_2^2 = x_1^2 + x_2^2 + x_3^2$$

Exercise 3. *Draw a diagram in* $\mathbb{R}^3$ *justifying Exercise 2 and verify that the length of the vector* $(4, 3, 12)$ *is* 13.

Theorem 2.20 (Cauchy-Schwarz inequality)**.** *In an inner product space* $(V, +, \cdot, \langle \cdot, \cdot \rangle)$, *we have*

$$\|\boldsymbol{u}\| \|\boldsymbol{v}\| \geq |\langle \boldsymbol{u}, \boldsymbol{v} \rangle|$$

for any $\boldsymbol{u}, \boldsymbol{v} \in V$.

Proof. Note that if either $\mathbf{u} = 0$ or $\mathbf{y} = 0$, then $|\langle \mathbf{u}, \mathbf{v} \rangle| = \|\mathbf{u}\| \|\mathbf{v}\|$. Therefore, we assume that both $\mathbf{u} \neq \mathbf{0}$ and $\mathbf{v} \neq \mathbf{0}$.

Define the vector $\alpha \mathbf{u} + \beta \mathbf{y}$, where $\alpha = \|\mathbf{v}\|$ and $\beta = \|\ \mathbf{u}\|$. Then, we have,

$$\begin{aligned} \|\alpha \mathbf{u} + \beta \mathbf{v}\|^2 &= (\|\mathbf{v}\| \mathbf{u} + \|\mathbf{u}\| \mathbf{v}) \cdot (\|\mathbf{v}\| \mathbf{u} + \|\mathbf{u}\| \mathbf{v}) \\ &= \|\mathbf{v}\|^2 \mathbf{u} \cdot \mathbf{u} + 2\|\mathbf{v}\| \|\mathbf{u}\| \mathbf{u} \cdot \mathbf{v} + \|\mathbf{u}\|^2 \mathbf{v} \cdot \mathbf{v} \\ &= 2\|\mathbf{v}\|^2 \|\mathbf{u}\|^2 + 2\|\mathbf{u}\| \|\mathbf{v}\| \mathbf{u} \cdot \mathbf{v} \\ &= 2\|\mathbf{u}\| \|\mathbf{v}\| (\|\mathbf{u}\| \|\mathbf{v}\| + \mathbf{u} \cdot \mathbf{v}) \end{aligned}$$

Since $\|\alpha \mathbf{u} + \beta \mathbf{v}\|^2$, $\|\mathbf{u}\|$ and $\|\mathbf{v}\|$ are all non-negative, we have $\|\mathbf{u}\| \|\mathbf{v}\| + \mathbf{u} \cdot \mathbf{v} \geq 0$, i.e., $\|\mathbf{u}\| \|\mathbf{v}\| \geq -\mathbf{u} \cdot \mathbf{v}$. Now, we consider the vector $\alpha \mathbf{u} + \beta \mathbf{v}$, where $\alpha = \|\mathbf{v}\|$ and $\beta = -\|\mathbf{u}\|$. Using the similar arguments, we have $\|\mathbf{u}\|\|\mathbf{v}\| \geq \mathbf{u} \cdot \mathbf{v}$. Putting both, we have

$$\|\mathbf{u}\|\|\mathbf{v}\| \geq \max\{\mathbf{u} \cdot \mathbf{v}, -\mathbf{u} \cdot \mathbf{v}\} = |\mathbf{u} \cdot \mathbf{v}|.$$

□

Exercise 4. *The Cauchy-Schwarz inequality is an equation, i.e.,* $|\langle \boldsymbol{u}, \boldsymbol{v}\rangle| = \|\boldsymbol{u}\|\|\boldsymbol{v}\|$ *if and only if one of the vectors* $\boldsymbol{u}$ *or* $\boldsymbol{v}$ *is a scalar multiple of the other.*

Definition 2.21. Let $(V, +, \cdot)$ be a vector space. The 4-tuple $(V, +, \cdot, \|\cdot\|)$ is a **normed vector space** iff the norm $\|\cdot\| : V \to \mathbb{R}_+$ satisfies the following properties:

(1) $\|\mathbf{v}\| = 0$ iff $\mathbf{v} = \mathbf{0}$, for any $\mathbf{v} \in V$,

(2) $\|\lambda \mathbf{v}\| = |\lambda|\|\mathbf{v}\|$, for any $\lambda \in \mathbb{R}$ and $\mathbf{v} \in V$, and

(3) Triangle inequality: $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$, for any $\mathbf{u}, \mathbf{v} \in V$.

Clearly, if $(V, +, \cdot, \langle \cdot, \cdot \rangle)$ is an inner product space, and we define the norm $\|\cdot\| : V \to \mathbb{R}_+$ by $\|\mathbf{v}\| := \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$, then $(V, +, \cdot, \|\cdot\|)$ is a normed vector space. Especially, the triangle inequality is a corollary of Cauchy-Schwarz inequality (as in Exercise). Therefore, normed vector spaces have less structures and are more general than inner product spaces.

Exercise 5. *Prove the triangle inequality using Cauchy-Schwarz inequality.*

Exercise 6. *Prove* $\|\boldsymbol{u} + \boldsymbol{v}\| = \|\boldsymbol{u}\| + \|\boldsymbol{v}\|$ *if and only if one of the vectors* $\boldsymbol{u}$ *or* $\boldsymbol{v}$ *is a non-negative scalar multiple of the other.*

In a normed vector space, the norm also induces a metric $d(\mathbf{u}, \mathbf{v}) := \|\mathbf{u} - \mathbf{v}\|$, which is called **Euclidean distance**. Note that a metric on V is a function $d : V \times V \to \mathbb{R}_+$ that satisfies the properties of the distance function in Proposition 2.17. The pair (V, d) is called a metric space (as in Proposition 2.17).

We can verify that this is indeed a metric on V. Clearly, $d(\mathbf{u}, \mathbf{v}) = 0$ iff $\mathbf{u} = \mathbf{v}$ because of property (1) of the norm; $d(\mathbf{u}, \mathbf{v}) = d(\mathbf{v}, \mathbf{u})$ and because of property (2) of the norm. Finally, this metric induced by the norm satisfies the triangle inequality because the norm satisfies the triangle inequality:

$$d(\mathbf{x}, \mathbf{y}) + d(\mathbf{y}, \mathbf{z}) = \|\mathbf{x} - \mathbf{y}\| + \|\mathbf{y} - \mathbf{z}\| \geq \|(\mathbf{x} - \mathbf{y}) + (\mathbf{y} - \mathbf{z})\| = \|\mathbf{x} - \mathbf{z}\| = d(\mathbf{x}, \mathbf{z}).$$

Therefore, a normed vector space is automatically a metric space. To summarize, in a vector space, an inner product induces a norm, which in turn induces a metric. If we consider $\mathbb{R}^n$ endowed with the dot product as an inner product space, then the dot product induces L_2 norm, which in turn induces the Euclidean distance

$$d_2(\mathbf{x}, \mathbf{y}) \coloneqq \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}.$$

Recall that we have shown that a valid inner product induces a valid norm, which in turn induces a valid metric, Because the dot product on $\mathbb{R}^n$ is a valid inner product, the Euclidean distance function d_2 is a valid metric; especially, it satisfies the triangle inequality required for metrics. (But, a metric space may have no vector structure - i.e., it may not be a vector space - so the concept of a metric space is a generalization of the concept of a normed vector space.)

In each of the following examples, you should verify that d is a metric by verifying that it satisfies each of the four conditions in Proposition 2.17 (1) - (4). For the norms on $\mathbb{R}^n$, you should draw the set of all points in $\mathbb{R}^2$ that satisfy $\|x\| = 1$.

Example. In $\mathbb{R}^n$ define $\|x\|_\infty \coloneqq \max\{|x_1|, \cdots, |x_n|\}$ and define $d : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}_+$ as $d(x, x') = \|x - x'\|_\infty = \max\{|x_1 - x_1'|, \cdots, |x_n - x_n'|\}$. This is called the *max*-norm or *sup*-norm.

Example. In $\mathbb{R}^n$ define $\|x\|_1 \coloneqq \sum_{i=1}^{n} |x_i|$ and $d(x, x') = \|x - x'\|_1 = \sum_{i=1}^{n} |x_i - x_i'|$. This is called the city block, Manhattan, or taxicab norm.

Example. Let p be a real number satisfying $p \geq 1$ and define $\|x\|_p$ on $\mathbb{R}^n_+$ by

$$\|x\|_p \coloneqq (|x_1|^p + |x_2|^p + \cdots + |x_n|^p)^{1/p}$$

This is called the ℓ_p norm. Note that the Euclidean norm is the ℓ_2-norm, the city block norm is the ℓ_1-norm, and the sup-norm is the ℓ_∞-norm.

Remark. If (X, d) is a metric space and S is a subset of X, then (S, d) is a metric space.

3 Matrices

3.1 Definition of Matrix

Definition 3.1. An $m \times n$ **matrix** is an array with $m \geq 1$ **rows** and $n \geq 1$ **columns**:

$$A = (a_{ij})_{ij} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

where the number a_{ij} is the i-th row and j-th column is called the ij-th **entry** or the ij-th **component**.

The set of matrices of size $m \times n$ is noted $\mathcal{M}_{mn}$. We note A_i the i-th row of A and A^j the j-th column of A, so that:

$$A = (A^1 \cdots A^n) = \begin{pmatrix} A_1 \\ \vdots \\ A_m \end{pmatrix}$$

Some particular matrices:

- Real numbers can be seen as a 1×1 matrix.
- A vector of $\mathbb{R}^k$ can be seen as a $k \times 1$ matrix (a column vector) or a matrix of size $1 \times k$ (a row vector).
- Matrices with as many rows as columns $m = n$ are called **square matrices**.
- The **zero matrix** of $\mathcal{M}_{mn}$ is the matrix with all entries equal to zero.
- A square matrix A is **diagonal** if all its non-diagonal elements are zero: $a_{ij} = 0$ for all i, j such that $i \neq j$. We note $A = \text{diag}(a_{11}, \cdots, a_{nn})$.
- The **unit matrix** of size n is the square matrix of size n having all its components equal to zero except the diagonal components, equal to 1. Denoted as I_n.
- A square matrix A is **upper-triangular** if all its elements below its diagonal are zeros: $a_{ij} = 0$ for all $i > j$.
- A square matrix A is **lower-triangular** if all its elements above its diagonal are zeros: $a_{ij} = 0$ for all $i < j$.

3.2 Operations on matrices

Definition 3.2. Let $A = (a_{ij})$ and $B = (b_{ij})$ be two $m \times n$ matrices, and $\lambda \in \mathbb{R}$.

(1) The sum $A + B$ is the matrix whose ij-th entry is $a_{ij} + b_{ij}$.

(2) The scalar multiplication of A by λ, λA is the matrix whose ij-th entry is λa_{ij}.

The matrix multiplication is defined over matrices of different sizes, although sizes need to be **conformable**: the product AB is only defined for matrices such that the number of columns of A is equal to the number of rows of B.

Definition 3.3. Let $A = (a_{ij})$ be a $m \times n$ matrix and $B = (b_{ij})$ be a $n \times s$ matrix. Their product AB is $m \times s$ matrix whose ij-th entry is:

$$(AB)_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}.$$

In fact, matrix multiplication should be interpreted as a linear mapping. Recall that in an $m \times n$ matrix with components in $\mathbb{R}$, each column can be viewed as a vector in $\mathbb{R}^m$ and each row can be viewed as a vector in $\mathbb{R}^n$. If we left-multiply matrix A by another matrix C, each row of the product CA is a linear combination of the rows of A; therefore, left-multiplying a matrix can be interpreted as a row transformation. If we right-multiply A by another matrix C, each column of the product AC is a linear combination of the columns of A; therefore, right-multiplying a matrix should be interpreted as a column transformation.

Proposition 3.4. Let A be a $m \times n$ matrix.

- The unit matrix is the neutral element of matrix multiplication: if A is $m \times n$, then $I_m A = A I_n = A$.
- The zero matrix is **absorbent**: $A0 = 0A = 0$.
- The multiplication is distributive w.r.t. the addition: $A(B + C) = AB + AC$ and $(B + C)A = BA + CA$.
- The multiplication is associative: $A(BC) = (AB)C$.
- $A(\lambda B) = \lambda(AB)$.

Contrary to the multiplication on real numbers, the matrix multiplication is in general NOT commutative: in general $AB \neq BA$.

Exercise 7. *Find the example where $AB \neq BA$. (Let A and B be 2×2 matrix.)*

Definition 3.5. A square matrix of size n is **invertible** (or **non-singular**) iff there exists a matrix B such that $AB = BA = I_n$. If such matrix exists, the inverse is unique and is denoted as A^{-1}.

Exercise 8. *Show that the inverse matrix A^{-1} is unique.*

Proposition 3.6. If $A, B \in \mathcal{M}_{nn}$ are invertible, then so is their product AB and $(AB)^{-1} = B^{-1}A^{-1}$.

For square matrices, it is also possible to define the **repeated products** or **powers** of a square matrix A.

Definition 3.7. $A^k = A \cdots A$. By definition, $A^0 = I_n$.

We say that a matrix A is **idempotent** if $A^2 = A$. We say that a matrix A is **nilpotent** if $A^k = 0$ for some integer k.

Exercise 9. *Find one example for idempotent matrix. (Keep your example with 2×2 size of matrix.)*

Definition 3.8. Let $A = (a_{ij})$ be a matrix. The **transpose** A' (or A^T) of A is the matrix obtained by changing its rows into its columns (and vice versa): $A^T = (a_{ji})$.

Obviously, if we apply the transpose operator twice, we end up back on $A : (A^T)^T = A$. Note that a row vector is the transpose of a column vector. The following properties of the transpose are easy to verify from the definition.

Proposition 3.9.

- $(\lambda A)^T = \lambda A^T$.
- Transpose of the sum: $(A + B)^T = A^T + B^T$.
- Transpose of the product: $(AB)^T = B^T A^T$.
- $(A^{-1})^T = (A^T)^{-1}$ provided that A^{-1} exists.

Definition 3.10. A square matrix A is **symmetric** iff it is equal to its transpose $A' = A$.

If A is a $m \times n$ real matrix, we can see its n columns $A_1, \cdots, A_n$ as n vectors of $\mathbb{R}^m$. Conversely, if $A_1, \cdots, A_n$ are n vectors of $\mathbb{R}^m$, we can see them as $m \times n$ matrix whose columns are the A_j. For instance, note this very useful way to write a linear combination of the vectors A_j using matrix multiplication:

$$\lambda_1 A_1 + \cdots + \lambda_n A_n = A\lambda, \text{ where } \lambda = \begin{pmatrix} \lambda_1 \\ \cdots \\ \lambda_n \end{pmatrix}$$

Since we can look at matrices as a family of vectors, we can also consider the vector space that these vectors span:

Definition 3.11. If $A = (A_1, \cdots, A_n)$ is a $m \times n$ matrix, we call the space $span(A_1, \cdots, A_n)$ spanned by the columns of A the **column space** (or **image**, noted $Im(A)$) of the matrix A.

Definition 3.12. The **column rank of a matrix** A is the rank of its column space $rank(A_1, \cdots, A_n)$.

We can do with rows as what we did with columns. We can see the n rows of a $m \times n$ matrix as m vectors of $\mathbb{R}^n$. We call the space $span(A_1, \cdots, A_m)$ spanned by the row vectors of A the **row space** of matrix A and its rank the **row rank** of the matrix. However, the row space is not as much useful as the column space, and:

Proposition 3.13. The row rank of a matrix is equal to its column rank.

Because the rows of the product matrix AB are linear combinations of rows of B, the basis of the rows of B can represent rows of AB and so we have $rank(AB) \leq rank(B)$. Because the columns of AB are linear combinations of columns of A, the basis of columns of A can represent columns of AB and so we have $rank(AB) \leq rank(A)$. As a result, we always have

$$rank(AB) \leq \min\{rank(A), rank(B)\}.$$

A consequence is that the rank of a $m \times n$ matrix is always smaller than both n and m.

Proposition 3.14. A square matrix A of size n is invertible if and only if $rank(A) = n$.

Proof. ($\Leftarrow$): Assume that $rank(A) = n$. Then, all vectors of $\mathbb{R}^n$ can be expressed as a linear combination of the columns of A. In particular, let the vectors e_j, $j = 1, \cdots, n$ of the canonical basis of $\mathbb{R}^n$. For all j, there exists a column vector $B^j \in \mathbb{R}^n$ such that $e_j = AB^j$. Denote $B = (B^1, \cdots, B^n)$ and $I_n = AB$. To prove that B is the inverse of A, we need to show that $BA = I_n$. Note that A' also has rank n. Using the same reasoning, there exists C such that $A'C = I_n$. Taking transpose, we have $C'A = I_n$. Then, $BA = (C'A)(BA) = C'(AB)A = C'A = I_n$.

($\Rightarrow$): Assume that A is invertible. We want to show that the columns of A are linearly independent. Consider a linear combination of the columns of A, $A\lambda$ for some vector $\lambda \in \mathbb{R}^n$ that is equal to zero, i.e., $A\lambda = 0$. Then, we have $\lambda = A^{-1}0 = 0$. $\square$

Definition 3.15. Let $A = (a_{ij})$ be a square matrix of size n. The **trace** of A, denoted as $tr(A)$ is:

$$tr(A) = \sum_{i=1}^{n} a_{ii}.$$

Proposition 3.16.

- The trace is linear: $tr(\lambda A) = \lambda tr(A)$ and $tr(A+B) = tr(A)+tr(B)$.
- A matrix and its transpose have the same trace: $tr(A^T) = tr(A)$.
- If AB and BA are square (but not necessarily A and B), $tr(AB) = tr(BA)$.

Definition 3.17. For a square matrix A, its **determinant**, denoted as $\det(A)$, is an element defined inductively in the following way:

(1) For a 1×1 matrix $A = a_{11}$, define its determinant as $\det(A) := a_{11}$.

(2) For a $n \times n$ matrix when $n \geq 2$, define its determinant as

$$\det(A) \equiv \sum_{j=1}^{n} (-1)^{1+j} a_{1j} \det(A_{-1,-j})$$

where $A_{-i,-j}$ is the matrix A with the i-th row and j-th column eliminated.

According to the definition above, for a 2×2 matrix

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

the determinant is $\det(A) = a_{11}a_{22} - a_{12}a_{21}$.

For a 3×3 matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

the determinant is

$$\begin{aligned}\det(A) &= a_{11}\det\left(\begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33}\end{bmatrix}\right) - a_{12}\det\left(\begin{bmatrix} a_{21} & a_{23} \\ a_{31} & a_{33}\end{bmatrix}\right) + a_{13}\det\left(\begin{bmatrix} a_{21} & a_{22} \\ a_{31} & a_{32}\end{bmatrix}\right) \\ &= a_{11}a_{22}a_{33} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31}\end{aligned}$$

Theorem 3.18. *Let n be an integer and consider the determinant function, taking n vectors of $\mathbb{R}^n$ as argument* $\det : \mathbb{R}^n \times \cdots \mathbb{R}^n \mapsto \mathbb{R}$

(1) $\det(I_n) = 1$.

(2) The determinant of a triangular (this includes diagonal) matrix $A = (a_{ij})$ is the product of its diagonal elements $\det(A) = \prod_{i=1}^{n} a_{ii}$.

(3) ***Multilinearity****:* $\det$ *is linear with respect to each of its argument:* $\forall k = 1, \cdots, n$,

$$\begin{aligned}\det(A_1, \cdots, \lambda A_k + \mu A'_k, \cdots, A_n) = \lambda \det(A_1, \cdots, A_k, \cdots, A_n) \\ + \mu \det(A_1, \cdots, A'_k, \cdots, A_n)\end{aligned}$$

(4) If any two columns of A are equal, then $\det(A) = 0$.

(5) ***Antisymmetry****: If two columns of A are interchanged, then the determinant changes by a sign.*

(6) If one adds a scalar multiple of one column to another, then the determinant does not change.

Theorem 3.19. *(1) A matrix and its transpose have the same determinant:* $\det(A') = \det(A)$. *(Equivalently, n vectors $A_1, \cdots, A_n$ of $\mathbb{R}^n$ are linearly independent if and only if* $\det(A_1, \cdots, A_n) \neq 0$.*)*

(2) A square matrix A is invertible if and only if $\det(A) \neq 0$.

(3) For two $n \times n$ matrices, we have $\det(AB) = \det(A)\det(B)$.

The theorem above implies that $\det(A^{-1}) = (\det(A))^{-1}$ for an invertible matrix A.

Exercise 10. *Show that* $\det(A^{-1}) = (\det(A))^{-1}$ *for an arbitrary 2×2 matrix for A.*

4 Systems of Linear Equations

In the economic theory, we encounter many variables and the relationship of these variables can be represented in a linear equation. The following shows the general linear system of m equations in n unknowns: :

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n & = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n & = b_2 \\ & \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n & = b_m \end{cases}$$

where a_{ij}'s and b_i's are all elements of $\mathbb{R}$ and the unknowns $x_1, \cdots, x_n$ also take values in $\mathbb{R}$. a_{ij} is the **coefficient** of the unknown x_j in the i-th equation. A **solution** of above system is an n-tuple of x's which satisfies m equations.

We can write the system of linear equations in a compact way by

$$A\mathbf{x} = \mathbf{b}$$

where $A = (a_{ij})$ is the $m \times n$ matrix, $\mathbf{b} = (b_1, \cdots, b_m)^T$ and $\mathbf{x} = (x_1, \cdots, x_n)^T$. A linear system of equations must have either no solution, one solution, or infinitely many solutions.

If we view $\mathbf{x}$ as a column transformation of the columns of A, then the equation asks us to find ways to represent the vector $\mathbf{b} \in \mathbb{R}^m$ as a linear combination of the columns of A. To clearly see this, write matrix A as $A = (\mathbf{a}_1, \cdots, \mathbf{a}_n)$, where $\mathbf{a}_i$ is the i-th column of A, we have $(\mathbf{a}_1, \cdots, \mathbf{a}_n)\mathbf{x} = (\mathbf{a}_1, \cdots, \mathbf{a}_n)(x_1, \cdots, x_n)^T = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n$. The following proposition shows the condition that tells us how many solution we have in a system.

Proposition 4.1. When the system has a solution,

(1) the solution is unique if the columns of A are linearly independent, i.e. $rank(A) = n$,

(2) the system has infinitely many solutions if $rank(A) < n$.

To see this, suppose $rank(A) = n$, then the columns of A are linearly independent and therefore, their linear representation of $\mathbf{b}$ is unique. On the other hand, suppose $rank(A) < n$, then the columns of A are linearly dependent. Therefore, they have a non-trivial linear representation of the zero vector, i.e. there exists $F \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ s.t. $A\mathbf{z} = \mathbf{0}$. Then, if $\mathbf{x}^*$ is a solution to the system, then $\mathbf{x}^* + \lambda\mathbf{z}$ is also a solution, for any $\lambda \in \mathbb{R}$ and so the solution is not unique.

Proposition 4.2 (General solution of a linear equation)**.** Let a vector $\mathbf{x}^*$ satisfy the equation $A\mathbf{x}^* = \mathbf{b}$ and let $H := \{\mathbf{z} : A\mathbf{z} = \mathbf{0}\}$. Then, the set $\{\mathbf{x} = \mathbf{x}^* + \mathbf{x}_h : \mathbf{x}_h \in H\}$ is the set of all solutions of the equation $A\mathbf{x} = \mathbf{b}$.

When $m = n$ and the square matrix A is invertible, the unique solution is clearly $\mathbf{x}^* = A^{-1}\mathbf{b}$. We also have an explicit formula for $\mathbf{x}^* = A^{-1}\mathbf{b}$, which is known as Cramer's rule.

Definition 4.3. For any $n \times n$ matrix A, let C_{ij} denote the (i, j)-th cofactor of A, that is, $(-1)^{i+j}$ times the determinant of the submatrix obtained by deleting row i and column j from A. The $n \times n$ matrix whose (i, j)-th entry is C_{ji}, the (j, i)-th cofactor of A, is called the **adjoint** of A and is written $\mathbf{adj}A$.

Theorem 4.4. *Let A be a nonsingular matrix, then*

$$A^{-1} = \frac{1}{\det A} adjA.$$

Theorem 4.5 (Cramer's Rule)**.** *Let A be a $n \times n$ invertible matrix and $\boldsymbol{b}$ be a $n \times 1$ column vector. The i-th entry of the $n \times 1$ column vector $x^* := A^{-1}\boldsymbol{b}$ can be calculated as*

$$x_i^* = \frac{\det(A_i)}{\det(A)}$$

for each i, where A_i is the $n \times n$ matrix formed by replacing the i-th column of A by $\boldsymbol{b}$ and leaving the other columns unchanged.

However, calculating the determinants is numerically difficult when the size of the matrices is large, since the determinant of $n \times n$ matrix has $n!$ terms.

Example1: Use Theorem 4.4 to invert the matrix,

$$A = \begin{bmatrix} 2 & 4 & 5 \\ 0 & 3 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Computing the cofactors,

$$C_{11} = + \begin{vmatrix} 3 & 0 \\ 0 & 1 \end{vmatrix} = 3,\ C_{12} = - \begin{vmatrix} 0 & 0 \\ 1 & 1 \end{vmatrix} = 0,\ C_{13} = + \begin{vmatrix} 0 & 3 \\ 1 & 0 \end{vmatrix} = -3,$$
$$C_{21} = - \begin{vmatrix} 4 & 5 \\ 0 & 1 \end{vmatrix} = -4,\ C_{22} = + \begin{vmatrix} 2 & 5 \\ 1 & 1 \end{vmatrix} = -3,\ C_{23} = - \begin{vmatrix} 2 & 4 \\ 1 & 0 \end{vmatrix} = 4,$$
$$C_{31} = + \begin{vmatrix} 4 & 5 \\ 3 & 0 \end{vmatrix} = -15,\ C_{32} = - \begin{vmatrix} 2 & 5 \\ 0 & 0 \end{vmatrix} = 0,\ C_{33} = + \begin{vmatrix} 2 & 4 \\ 0 & 3 \end{vmatrix} = 6,$$

and $\det A = -9$, and

$$\text{adj}A = \begin{pmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{23} & C_{33} \end{pmatrix} = \begin{pmatrix} 3 & -4 & -15 \\ 0 & -3 & 0 \\ -3 & 4 & 6 \end{pmatrix}$$

So,

$$A^{-1} = -\frac{1}{9} \begin{pmatrix} 3 & -4 & -15 \\ 0 & -3 & 0 \\ -3 & 4 & 6 \end{pmatrix}$$

Example2: Use Cramer's rule to calculate x_3 under the following equation:

$$\begin{pmatrix} 1 & 1 & 1 \\ 12 & 2 & -3 \\ 3 & 4 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 5 \\ -4 \end{pmatrix}.$$

The determinant of A is 35. The determinant of

$$B_3 = \begin{pmatrix} 1 & 1 & 0 \\ 12 & 2 & 5 \\ 3 & 4 & -4 \end{pmatrix}$$

is also 35. Thus, $x_3 = |B_3|/|A| = 1$.

Exercise 11 (SB 9.13)**.** *Use Cramer's rule to solve the following systems of equations:*

1.

$$\begin{aligned} 5x_1 + x_2 &= 3 \\ 2x_1 - x_2 &= 4 \end{aligned}$$

2.

$$\begin{aligned} 2x_1 - 3x_2 &= 2 \\ 4x_1 - 6x_2 + x_3 &= 7 \\ x_1 + 10x_2 &= 1 \end{aligned}$$